Review notes · for you

Data detectives: unit check

Data science · Level 1 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- The data science loop
- Middles and spread
- Charts that fit and tell the truth
- Careful claims and private pupils

Activity: Revise in pairs

1. Pairs (2 min): Work with a partner and share one sheet for your answers.
2. Quiz (15 min): Read each question, agree on an answer and say why.
3. Check (8 min): Compare with the answers and redo the ones you got wrong.

Discussion

- Which idea from this unit will you use first?
- Which school record will you explore with your class?
- What will you change about sharing results?

Check yourself

- One habit I will use this week, and where.

Try it this week

Teach one idea from this unit to someone at home, using a chart you draw yourself.

Quiz sheet

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Name _____ Class _____ Date _____

1 **A good data question is one you can count or measure.**

◯ True ◯ False

2 **What is the mean of 2, 4 and 6?**

◯ 6 ◯ 12 ◯ 4

3 **Use 3, 7, 7, 9, 14.**

The mode is _____ and the range is _____.

Word bank: 9 · 7 · 11 · 14

4 **Match each question to a chart.**

Favourite colours in our class	• • • •	Line chart
Rain each day this month	• • • •	Bar chart

5 **A bar chart's scale starts at 90 instead of 0. What happens?**

◯ Small differences look huge ◯ The bars disappear ◯ Nothing changes

6 **One very big number can pull the mean up.**

◯ True ◯ False

7 **Put the loop in order.**

☐ Collect

☐ Ask

☐ Share

☐ Look

8 **Which are private? Choose all that apply.**

☐ The number of pupils in the school ☐ Your home address ☐ Your test marks

9 **"Most people love our sweets!" They asked 5 people at their shop. What is the problem?**

◯ Nothing ◯ Too few people, and only their customers ◯ Sweets are healthy

10 **Every chart needs a title and labels.**

◯ True ◯ False

Check your answers

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1 **True**

If you can count or measure it, data can answer it.

2 **4**

2 + 4 + 6 = 12, and 12 ÷ 3 = 4.

3 **The mode is 7 and the range is 11.**

7 appears twice, and 14 − 3 = 11.

4 **Favourite colours in our class — Bar chart**
Rain each day this month — Line chart

Compare groups with bars, show change with a line.

5 **Small differences look huge**

Cutting the axis exaggerates differences.

6 **True**

The median stays in the middle, but the mean moves.

7 **Top to bottom: 2, 1, 4, 3**

Ask, collect, look, share, then ask again.

8 **Your test marks; Your home address**

The school's total belongs to nobody in particular.

9 **Too few people, and only their customers**

A small group of fans can't speak for everyone.

10 **True**

Without them, people can't tell what the chart shows.